MSG No. 131.437 $P4_2/m'mc$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m'mm`.

No.	position	mapping
1	$[0, 0, 0]$	`[1,4,5,6,9,10,13,16]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,3,7,8,11,12,14,15]`

Table 2: Wyckoff site: `2b`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,4,5,6,9,10,13,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,7,8,11,12,14,15]`

Table 3: Wyckoff site: `2c`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,6,9,10,13,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,7,8,11,12,14,15]`

Table 4: Wyckoff site: `2d`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,6,9,10,13,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,7,8,11,12,14,15]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,5,6,11,12,14,15]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,7,8,9,10,13,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,4,5,6,11,12,14,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,7,8,9,10,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2mm`.

No.	position	mapping
1	$[0, 0, z]$	`[1,4,5,6]`
2	$\left[0, 0, z + \frac{1}{2}\right]$	`[2,3,7,8]`

continued ...

Table 7

No.	position	mapping
3	$[0,\ 0,\ -z]$	`[9,10,13,16]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[11,12,14,15]`

Table 8: Wyckoff site: `4h`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,10,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,12,14,15]`

Table 9: Wyckoff site: `4i`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,10,13,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11,12,14,15]`

Table 10: Wyckoff site: `4j`, site symmetry: `m'2'm`.

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,6,9,16]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,8,11,15]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[3,7,12,14]`
4	$[-x,\ 0,\ 0]$	`[4,5,10,13]`

Table 11: Wyckoff site: `4k`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,6,9,16]`
2	$\left[\frac{1}{2},\ x,\ 0\right]$	`[2,8,11,15]`
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[3,7,12,14]`
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,5,10,13]`

Table 12: Wyckoff site: `4l`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,6,9,16]`
2	$[0,\ x,\ 0]$	`[2,8,11,15]`
3	$[0,\ -x,\ 0]$	`[3,7,12,14]`

continued ...

Table 12

No.	position	mapping
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[4,5,10,13]

Table 13: Wyckoff site: 4m, site symmetry: m'2'm.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	[1,6,9,16]
2	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[2,8,11,15]
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[3,7,12,14]
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	[4,5,10,13]

Table 14: Wyckoff site: 8n, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	[1,11]
2	$\left[-x,\, x,\, \frac{3}{4}\right]$	[2,10]
3	$\left[x,\, -x,\, \frac{3}{4}\right]$	[3,9]
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[4,12]
5	$\left[-x,\, x,\, \frac{1}{4}\right]$	[5,15]
6	$\left[x,\, -x,\, \frac{1}{4}\right]$	[6,14]
7	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[7,13]
8	$\left[x,\, x,\, \frac{3}{4}\right]$	[8,16]

Table 15: Wyckoff site: 8o, site symmetry: .m.

No.	position	mapping
1	$[0,\, y,\, z]$	[1,5]
2	$\left[-y,\, 0,\, z+\frac{1}{2}\right]$	[2,7]
3	$\left[y,\, 0,\, z+\frac{1}{2}\right]$	[3,8]
4	$[0,\, -y,\, z]$	[4,6]
5	$[0,\, -y,\, -z]$	[9,13]
6	$[0,\, y,\, -z]$	[10,16]
7	$\left[y,\, 0,\, \frac{1}{2}-z\right]$	[11,14]
8	$\left[-y,\, 0,\, \frac{1}{2}-z\right]$	[12,15]

Table 16: Wyckoff site: 8p, site symmetry: .m.

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	[1,5]
2	$\left[-y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[2,7]
3	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[3,8]
4	$\left[\frac{1}{2},\, -y,\, z\right]$	[4,6]
5	$\left[\frac{1}{2},\, -y,\, -z\right]$	[9,13]

continued ...

Table 16

No.	position	mapping
6	$\left[\frac{1}{2}, y, -z\right]$	[10,16]
7	$\left[y, \frac{1}{2}, \frac{1}{2} - z\right]$	[11,14]
8	$\left[-y, \frac{1}{2}, \frac{1}{2} - z\right]$	[12,15]

Table 17: Wyckoff site: 8q, site symmetry: m'..

No.	position	mapping
1	$[x, y, 0]$	[1,16]
2	$\left[-y, x, \frac{1}{2}\right]$	[2,15]
3	$\left[y, -x, \frac{1}{2}\right]$	[3,14]
4	$[-x, -y, 0]$	[4,13]
5	$[-x, y, 0]$	[5,10]
6	$[x, -y, 0]$	[6,9]
7	$\left[-y, -x, \frac{1}{2}\right]$	[7,12]
8	$\left[y, x, \frac{1}{2}\right]$	[8,11]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[-y, x, z + \frac{1}{2}\right]$	[2]
3	$\left[y, -x, z + \frac{1}{2}\right]$	[3]
4	$[-x, -y, z]$	[4]
5	$[-x, y, z]$	[5]
6	$[x, -y, z]$	[6]
7	$\left[-y, -x, z + \frac{1}{2}\right]$	[7]
8	$\left[y, x, z + \frac{1}{2}\right]$	[8]
9	$[x, -y, -z]$	[9]
10	$[-x, y, -z]$	[10]
11	$\left[y, x, \frac{1}{2} - z\right]$	[11]
12	$\left[-y, -x, \frac{1}{2} - z\right]$	[12]
13	$[-x, -y, -z]$	[13]
14	$\left[y, -x, \frac{1}{2} - z\right]$	[14]
15	$\left[-y, x, \frac{1}{2} - z\right]$	[15]
16	$[x, y, -z]$	[16]